MILITARY SPECIFICATION

RUBBER FABRICATED PRODUCTS FOR M17A1/M17A2 MASKS

This specification is approved for use by US Army Armament, Munitions and Chemical Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers fabricated products of vulcanized rubber, synthetic rubber, or rubber-like compositions alone or in combination.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-283 - Water Purification Tablet, Iodine.

STANDARDS

FEDERAL

Fed. Std. No. 123 - Marking for Shipment (Civil Agencies).
Fed. Std. No. 601 - Rubber: Sampling and Testing.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:55Z
Check the source to verify that this is the current version before use.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-289 - Visual Inspection Guide for Rubber Sheet Material.
MIL-STD-298 - Visual Inspection Guide for Rubber Extruded Goods.
MIL-STD-407 - Visual Inspection Guide for Rubber Molded Items.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

US ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH AND DEVELOPMENT CENTER

B5-81-102	Gasket.
C5-81-103	Gasket.
C5-81-124	Tube.
B5-81-126	Tube.
C5-81-127	Mouthpiece.
C5-81-130	Tube.
C5-81-132	Tub e.
E5-81-133	Mouthpiece.
B5-81-163	Gasket.
E136-13-1025-1	Flexure Tester, Cyclic, Q181, Assembly.
D136-19-14	Puncture Tester, Natural Rubber, Q38, Assembly.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

CODE OF FEDERAL REGULATIONS

21 CFR 10 to 199.

(Application for copies of this Code of Federal Regulations should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

Source: https://assist.dla.mil -- Downloaded: 202610-22T20:55Z
Check the source to verify that this is the current version before use.

ASTM STANDARDS

D395 - Compression Set of Vulcanized Rubber.
D412 - Tension Testing of Rubber.
D572 - Accelerated Aging of Vulcanized Rubber by the Oxygen-Pressure Method.
D573 - Accelerated Aging of Vulcanized Rubber by the Oven Method.
D624 - Tear Resistance of Vulcanized Rubber.
D746 - Brittleness Temperature of Plastics and Elastomers by Impact.
D792 - Specific Gravity and Density of Plastics by Displacement.
D797 - Young's Modulus in Flexure of Elastomers at Normal and Subnormal Temperatures.
D1149 - Accelerated Ozone Cracking of Vulcanized Rubber.
D2240 - Indentation Hardness of Rubber and Plastic by Means of a Durometer.

(Copies may be obtained from ASTM, 1916 Race Street, Philadelphia, PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Material. Rubber products finished under this specification shall be made from new rubber compositions of the types, grades, classes, and specific characteristics designated on the applicable part drawing or in tables I and 11, as applicable.

3.2 Toxicity. The rubber fabricated parts, except Drawing C5-8I-103, C5-81-132, and B5-81-163 shall be non-toxic when used in the chemical biological mask system. Objective evidence shall be submitted in accordance with 4.1.2.1. Once the rubber formulations have been approved by the Government , the contractor shall not make any change in the formulation or manufacturing processes.

3.3 Manufacture. The piece parts shall be manufactured to conform to the applicable drawings forming a part of this specification.

3.4 Tube (Drawing C5-81-124).

3.4.1 Flexure. The tube shall withstand 5,000 cycles minimum without evidence of failure, when tested as specified in 4.2.4.1*

3.4.2 Blocking. The tube shall not stick together (block), when tested as specified in 4.2.4.2.

3.4.3 Leakage. The tube shall not leak when pressurized to 1.5 inches of mercury and tested as specified in 4.2.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:55Z
Check the source to verify that this is the currentversion before use.

TABLE I. Ethylene propylene terpolymers, physical properties

Property	Requirement		Test method	Number of specimens
	Unaged	Aged*		
Tensile strength, MPa, min	10.3	-	ASTM D412	3
% change, max	-	-25	ASTM D573	3
Ultimate elongation, %, min	400		ASTM D412	3
% change, max	-	-25	ASTM D573	3
Durometer hardness	55 ± 5	-	ASTM D2240	1
change, max	-	+7		1
Compression set (22 hrs at 68°C)			ASTM D395	
% max	25			3
Ozone resistance (72 hr at 50°C) and 100 pphm ozone	No cracks or breaks		ASTM D1149	6
Low temperature brittleness at -55°C	Pass		ASTM D746	3
Tear resistance, kN/m, min, die C	21.0		ASTM D624	6
**Puncture resistance, psi, min	9.5		4.2.4.3	6
Specific gravity, max	1.12		ASTM D792	1

*70 hrs at 70°C
**Dwg B5-81-126 only

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

TABLE 11. Silicone rubber physical properties. (See Notes 1 and 2)

Property	Requirement		Fed Std 601 or ASTM test method	Number of specimens
	Unaged	Aged		
Tensile strength, MPa, min	4.83[A]/4.48[B]		ASTM D412	3
% change, max		-30[A]/-20[B]	ASTM D573	3
Ultimate elongation, % min	225[A]/100[B]		ASTM D412	3
% change, max		-50[A]/-40[B]	ASTM D573	3
Durometer, hardness, Shore A ± 5	50[A]/60[B]		ASTM D2240	1
Change, max		±15[A]/±10[B]	ASTM D2240	1
Compression set (70 hrs at 150°C) A (22 ± 1/4 hrs at 149° ± 3°C) B % max		35[A]/15[B]	ASTM D395 Method B	3
Resilience, % min	48[A]/51[B]		ASTM D945	3
Young's Modulus in flexure 24 hours at -75°C psi max		5000[A]	ASTM D797	2
Brittle point (shall not fail after single blow impact) min °C		-75[A]/-62[B]	ASTM D2137 Method A	3
Torsional stiffness ratio 72 hrs at -75°C, ratio max		15[A]	ASTM D1053	3
Tear resistance, kN/m, min	10.5[B]		ASTM D624	3
After water immersion for 70 hrs at 100°C change, % max		+5[B]	6211	3

A - Gaskets, B5-81-102
B - Tube, C5-81-124

NOTE 1: Before testing the silicone rubber test sample shall be placed in a temperature chamber at a temperature of -62 + 3°C for a minimum of 72 hours. After this 72 hour period, the temperature of the test samples shall be brought to equilibrium under standard conditions as defined in ASTM D1349 and then shall be tested for conformance with all the requirements in table II above in accordance with the test method specified"

NOTE 2: oven aging for tensile strength, elongation, and durometer hardness shall be 70 hours at 225 ± 3°C.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

3.5 Workmanship. The rubber products shall be free from grease, oil, foreign matter, and defects as specified in MIL-STD-289, MIL-STD-298, and MIL-STD-407, as applicable to the method of manufacture.

3.6 Oxygen aging. Molded parts shall show no increased hardness or tackiness when aged in an oxygen environment in accordance with ASTM D572 for 72 hours.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1, 3.2, and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.1.2.1 Toxicity. After date of award of contract, and within the time specified in the contract, and without additional cost to the Government, the contractor shall submit the information listed below which will constitute his evidence for establishing the non-toxicity of the rubber parts. The US Army Surgeon General shall act as advisor to the Contracting Officer in evaluating this data for determining if the formulation and parts are acceptable. The information relating to the description of the candidate formulations may be classified as "company confidential" in which case it will be so treated by the Government.

Information relating to candidate formulations:

(1) The name of the contractor and the producing subcontractor, if any l

(2) Description of the item. Copies of applicable specifications shall be included.

(3) The number of items under procurement for which the proposed formulation will be used.

(4) A description of the candidate formulation that shall include:

(a) The chemical and trade names of the ingredients.

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:55Z
Check the source to verify that this is the currentversion before use.

(b) The quantity of each ingredient, including the range of variation that can be tolerated without loss of quality in the finished item. Quantities shall be expressed in percent by weight and parts by weight.

(c) The proposed mold lubricants, dusting powders, wet spray or other materials used in the manufacturing process,

(d) proposed cure times and temperatures.

(5) Results of previous toxicity evaluations demonstrating acceptability or evidence of prior Government approval of the proposed formulations.

(6) Information and data to substantiate the non-toxicity of the formulation. This information shall include a complete description of any programs the contractor or his subcontractor have carried out to prove non-toxicity; evidence that the proposed formulation is the same as that used in his commercial products, and information on the nature, extent and length of time of his commercial experience with the formulation. Other objective evidence is the certification that the ingredients of the formulation conform to the Federal Food, Drug and Cosmetic Act Limitations on Food Additives and Extractives as specified in 21CFR121.2562 and 21CFR121.2589.

(7) The information shall include the following test data: The tests described below shall be run separately for each group of similar types of rubber as follows:

Group 1 - Silicone Rubber

Kink Tube	Dwg C5-81-124
Quick Disconnect Gasket	Dwg B5-81-102

Group 2 - EPT Rubber

Drink Mouthpiece	Dwg C5-81-127
Internal Resuscitation Tube	Dwg C5-81-130
External Drink Tube	Dwg B5-81-126
Mouthpiece	Dwg E5-81-133

The water contact surface of the rubber article in the finished form in which it is to be used, when extracted with distilled water at reflux temperature shall yield total extractives not to exceed 20 mg per square inch during the first seven hours of extraction, nor to exceed 1 mg per square inch during the succeeding two hours of extraction

(8) If any ingredient in the formulation for a part in group 1 or group 2 is not listed in 21CFR121.2562 and 21CFR121.2589 then a separate extractive test shall be run for that part and a qualitative and quantitative analysis made of the extractives obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

(9) Repeat the test procedure outline in (7) above using new material, except use a solution of distilled water containing one iodine water purification tablet, conforming to MIL-W-283, per quart of distilled water. The amount of total extractives shall not exceed that stated above when the test temperature is maintained at 160 ± 5°F.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the fabricated rubber products of the same dimensions and composition, produced by one manufacturer, at one plant, from the same materials and equipment, under essentially the same manufacturing conditions. However, no more than one batch of rubber (see 6.3) may be included in the same lot. A new lot shall be established following the occurrence of any event affecting the homogeneity of the lot or use of an alternative approved formulation.

4.2.2 Sampling.

4.2.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For defects 101, 102, 103, 104 and 105 in 4.2.3.1(c), level S-4 shall be used.

4.2.2.2 For destructive physical properties testing. Sixteen test slabs 6 inches by 6 inches by 0.075 ± 0.010 inch and five test buttons produced in accordance with the applicable test methods specified in this specification (for EPT components C5-81-103, B5-81-126, C5-81-127, C5-81-130, C5-81-132, B5-81-133, and B5-81-163), (for silicone rubber components B5-81-102 and C5-81-124) shall be submitted with each lot for testing. The test slabs and test buttons shall be made from stock used for the regular production lot and shall have a cure equivalent to that of the regular production lot. A randomly selected sample of 20 molded parts shall be submitted with each lot for oxygen aging testing.

4.2.3 Inspection procedures. The sample components shall be examined in accordance with the classification of defects and MIL-STD-105. The sample tubes (C5-81-124) shall be tested in accordance with 4.2.4.1, 4.2.4.2, and 4.2.4.4. Sample test slabs or buttons shall be tested in accordance with the specified applicable ASTM, Fed. Std. No. 601 test methods, and 4.2.4.3 of this specification as specified in tables I and II. The sample molded parts for oxygen aging testing shall be tested in accordance with ASTM D572. The acceptance number is zero.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

4.2.3*1 Classification of defects.

(a) Gasket (Drawing B5-81-102).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Outside diameter
102	Inside diameter
103	Workmanship (3.5)
104	Concentricity incorrect
105	Parallelism incorrect
Minor:	AQL 4.0 percent defective
201	Thickness incorrect
202	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

(b) Gasket (Drawing C5-81-103).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Hole center distance
102	Workmanship (3.5)
Minor :	AQL 4.0 percent defective
201	Inside diameter (0.21)
202	Inside diameter (0.323)
203	Thickness incorrect
204	Wall thickness
205	Radius (0.25)
206	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

(c) Tube (Drawing C5-81-124).

Categories	Defects	Acceptance standards
Major:	AQL 0.1 percent defective	
101	Flexure	4.2.4.1
102	Blocking	4.2.4.2
103	Leakage	4.2.4.4
	AQL 1.0 percent defective	
104	Inside diameter	
105	Wall thickness	
106	Workmanship (3.5)	
107	Damage such as tears, nicks, or jagged edges	
Minor:	AQL 4.0 percent defective	
201	Overall length	
202	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407	

(d) Tube (Drawing B5-81-126).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Wall thickness
102	Inside diameter
103	Workmanship (3.5)
Minor:	AQL 4.0 percent defective
201	Overall length
202	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

(e) Mouthpiece (Drawing C5-81-127).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Wall thickness (0.093)
102	Workmanship (3.5)
Minor:	AQL 4.0 percent defective
201	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

(f) Tube (Drawing C5-81-130).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Wall thickness
102	Inside diameter (0.44)
103	Maximum length (free state)
104	Workmanship (3.5)
Minor:	AQL 4.0 percent defective
201	Outside diameter (0.75)
202	Length dimension (0.31)
203	Loose soapstone powder
204	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

(g) Tube (Drawing C5-81-132).

Categories	Defects
Critical:	None defined
Major:	AQL 2.5 percent defective
101	Wall thickness
102	Inside diameter (0.62)
103	Maximum length (free state)
104	Workmanship (3.5)
105	Maximum length incorrect (extended under 3 lb. applied force)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

Minor: AQL 4.0 percent defective

201	Outside diameter (1.08)
202	Length dimension (0.50) (2 places)
203	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407
204	Loose soapstone powder

(h) Mouthpiece (Drawing E5-81-133).

Categories — Defects

Critical: None defined .

Major: AQL 2.5 percent defective

101	Workmanship (3.5)

Minor: AQL 4.0 percent defective

201	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

(i) Gasket (Drawing B5-81-163).

Categories — Defects

Critical: None defined

Major: AQL 2.5 percent defective

101	Wall thickness (0.12)
102	Inside diameter
103	Workmanship (3.5)

Minor: AQL 4.0 percent defective

201	Thickness
202	Minor defects listed in MIL-STD-289, MIL-STD-298, and MIL-STD-407

4.2.4 Tests.

4.2.4.1 Flexure. Tube flexure shall be determined with the Q181 Flexure Cyclic Tester Assembly (E136-13-1025-1) at a temperature of 70° ± 5°F with the tube bending back upon itself to an included angle of 75 ± 5 degrees. The tube shall be flexed to meet requirements of 3.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

4.2.4.2 Blocking. A full length of tubing shall be placed between two metal plates of l/4-inch minimum thickness. A 10-pound minimum load shall be applied to plates, causing tubing to collapse. The load shall be kept on plates for 8 hours minimum at 200° ± 10°F.

4.2.4.3 Puncture resistance. Resistance to hydrostatic pressure shall be determined with the Q38 Natural Rubber Puncture Tester Assembly (D136-19-14) using a 0.075 + 0.010 inch thick test slab.

4.2.4.4 Leakage. Upon completion of the flexure test specified in 4.2.4.1 the leakage of the same tube shall be determined by closing one end of the tube, pressurizing the tube per 3.4.3 and submerging the tube under water for a minimum of five seconds. A steady stream of bubbles shall be considered as evidence of leakage.

5. PACKAGING

5.1 Unit packing, packing, and marking, interplant shipment (see 6.4). The rubber fabricated products shall be unit packed and packed to provide adequate protection from physical damage from the supply source to the first receiving activity for immediate use or further processing. Shipping containers shall be in compliance with the rules and regulations applicable to the mode of transportation. Marking shall be in conformance with Fed. Std. No. 123 and shall include the contractor's name, address, and contract number.

6. NOTES

6.1 Intended use. This rubber products covered in this specification are intended for use in the M17A1 Field Chemical-Biological Mask and Ml resuscitation tube, and M17A2 field chemical-biological mask.

6.2 Ordering data. Acquisition documents should specify the title, number, and date of this specification.

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process *or* subjected to some physical mixing operation intended to make the final product substantially uniform.

6.4 Interplant shipment. Packaging for interplant shipment is for supplies and materials that do not enter the military supply system. Typical interplant shipments are shipments from a vendor to a subcontractor or a prime contractor, or from a subcontractor to a prime contractor, or from a vendor or contractor to a military arsenal or plant.

Custodians:

Army - EA

Preparing activity:

Army - EA

Project No. 4240-A784

Source: https://assist.dla.mil -- Downloaded: 2026l0-22T20:55Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY CARD

FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
Armament Research and Development Center
US Army Armament, Munitions and Chemical Command
ATTN: DRSMC-TSC-S(A)
Aberdeen Proving Ground, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-51283C(EA)	RUBBER FABRICATED PRODUCTS FOR M17A1/M17A2 MASKS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426

EDITION OF 1 OCT 76 IS OBSOLETE

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:55Z
Check the source to verify that this is the current version before use.